

Jerry Liu

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EDUCATION

University of Waterloo

BASc, Mechatronics Engineering

Waterloo, Ontario

2025 – 2030

- **GPA:** 3.88/4.0
- **Awards:** Arthur F. Church Entrance Scholarship (\$10,000) – Awarded to only 1 student in the Mechatronics cohort for outstanding academic and extracurricular achievement.

SKILLS

Programming Languages: Python, C++, Great Cow Basic, Processing, JavaScript, HTML, CSS, TypeScript

Technical Skills: SolidWorks, AutoCAD, PyTorch, Data preprocessing, Git/Github, Visual Studio Code, Microsoft Excel, Soldering / PCB Assembly, SQL, FastAPI, PostgreSQL, Next.js, React, Supabase, Gemini API

RELEVANT EXPERIENCE

Undergraduate Research Assistant

University of Waterloo

Waterloo, Ontario

Feb 2026 – Present

- Designing and building drone-based remote sensing platform for greenhouse gas detection using mid-infrared spectroscopy
- Developing multi-sensor fusion pipeline integrating LiDAR, GPS/IMU, and anemometer data for source localization
- Conducted literature review and contributed to component selection under Prof. Shunde Yin

Research Assistant

University of Waterloo

Waterloo, Ontario

August 2024 – June 2025

- Researched and selected multi-modal sensors for time-series degradation analysis on CNC milling machines
- Developed Python data processing pipelines for data trials, converting raw sensor data to usable PyTorch files
- Collected and processed high-frequency time-series data for remaining-useful-life prediction research using transformer models

Co-Founder, Logistics Lead

NeoDev League | [Link](#)

Waterloo, Ontario

May 2024 – June 2025

- Co-founded a non-profit organizing large-scale engineering competitions for high school students
- Led logistics for a 10-hour hackathon-style event, managing scheduling, facilities, and technical resources
- Secured \$10,000+ in sponsorships by contacting 500+ companies; reached 150 students across 10 schools

Team Member

First Robotics – 2702 Rebels

Waterloo, Ontario

October 2022 – October 2024

- Modelled robot subsystems in SolidWorks, ensuring structural integrity and manufacturability under competition constraints
- Prototyped components using industry-standard manufacturing tools, translating design concepts into functional assemblies
- Contributed to a positive, solution-oriented team culture that led to qualification for the World Championships

PROJECTS

Canopi: Rental Matchmaker | [Link](#)

March 2026

- Developed full-stack rental platform with Next.js, TypeScript, and Supabase, integrating a custom scraper to aggregate real-time housing data across Canada
- Implemented a custom compatibility algorithm to calculate user-to-property match scores and integrated the Gemini API to build an AI chat interface for inferring user preferences through natural language.

ArtistBlend: Automated Playlist Curator

Feb 2026

- Developed a full-stack web application using Next.js and TypeScript that leverages the Spotify Web API to dynamically filter and curate custom playlists from a user's library of liked songs
- Implemented OAuth 2.0 for secure user authentication and engineered logic to handle paginated API requests, enabling the efficient processing of thousands of tracks based on specific artist metadata

Neural Chess Engine | [Link](#)

Nov 2025

- Built and trained a 6-layer PyTorch neural network using a custom 12-channel (12×8×8) board encoding on millions of high-ELO games
- Implemented a minimax engine with alpha-beta pruning and deployed the model to Lichess via the lichess-bot API for real-time play

Firefighter Robot | [Video](#)

Feb 2025 – June 2025

- Programmed autonomous navigation and fire detection in GCBASIC, integrating wall sensors, phototransistors, and IR flame sensors for real-time control
- Designed and manufactured a custom PCB and 3D-printed chassis; assembled, soldered, and debugged the full system to validate end-to-end performance